Zirui Chen

Email: zc2445@columbia.edu
ziruichen@ucsb.edu
Phone: 929-624-9330
Mailing Address: 232 Forest Drive,
Goleta, CA, 93117

PhD candidate in computational astrophysics with expertise in stochastic processes, Markov analysis, probability theory and probability density function modeling, Python and C++ programming, quantitative data analysis, and large-scale parallelized computing.

Education & Professional Experience

Columbia University, NY, USA
B.A. in Physics and Astronomy (GPA 4.06/4.3) September 2018–May 2022
Advisors: Greg Bryan, Drummond Fielding, David Kipping
Awards & Honors: Summa Cum Laude, Departmental Honors (Physics), Phi Beta Kappa, John Jay Scholar, Dean's List

Center for Computational Astrophysics, Flatiron Institute, NY, USA
Guest Researcher May 2022–August 2022

University of California, Santa Barbara, CA, USA
M.A. and Ph.D. in Astrophysics (GPA 3.9/4.0) September 2022–Present
Advisor: S. Peng Oh
Awards & Honors: James and Mary Jo Hartle Graduate Fellowship

Skills

- Programming: Python; C++; Linux/Bash; MPI; OpenMP.
- Scientific Computing & Modeling: Stochastic Processes; Markov Analysis; Probability Theory; Monte Carlo Methods; Computational Fluid Simulations; Large-Scale Parallel Computing; Numerical Techniques for Solving Differential Equations.

Selected Projects & Publications

- **Stochastic Transition Matrix Modeling of Probability Density Functions in Turbulent Flows** (May 2026–Present)

  Modeling the probability density function of multi-phase, turbulent astrophysical gas using Markov analysis and stochastic transition matrices. Use eigenvectors and eigenvalues of transition matrices to quantify relaxation toward stationary distributions. Analyze deviations from standard diffusion in 3D simulations of turbulent flows, including Lévy flight and sub-diffusion due to trapping.

- **3D Hydrodynamic Simulations of Molecular Gas in Galactic Outflows** (September 2022–April 2024)

  Designed 3D hydrodynamic simulations using the C++ based framework Athena++ to study cold cloud survial in hot galactic outflows. Addressed the computational challenge of resolving gas spanning 5 orders of magnitude in temperature and density. Developed efficient numerical schemes for radiation self-shielding and dust evolution and performed high-resolution parallelized simulations on HPC clusters using 10k+ node-hours.

  Publication: **Zirui Chen** and S. Peng Oh, "The Survival and Entrainment of Molecules and Dust in Galactic Winds", 2024, Monthly Notices of the Royal Astronomical Society.

- **Markov Chain Prediction of Exoplanet Transits** (January 2019–July 2022)

  Designed N-body simulations of planets orbiting binary star systems via Python. Described exoplanet transit events using a Markov chain, which demonstrates memory in the system and provides probabilistic predictions of future transit events. Tested the Markov chain on millions of planetary systems generated through large-scale, parallelized simulation runs on HPC clusters.

Publication: **Zirui Chen** and David Kipping, "The Number of Transits Per Epoch for Transiting Misaligned Circumbinary Planets", 2022, Monthly Notices of the Royal Astronomical Society.

- **Analytic Model for Turbulent Mixing Layers Using Partial Differential Equations** (June 2020–June 2023)

  Derived partial differential equations for turbulent mixing layers at multi-phase gas interfaces and developed a novel parameterization of turbulence-driven conductivity and viscosity. Used numerical integration schemes in Python to reproduce solutions from large-scale 3D simulations in minutes instead of thousands of node-hours.

  Publication: **Zirui Chen**, Drummond Fielding, and Greg Bryan, "The Anatomy of a Turbulent Radiative Mixing Layer: Insights from an Analytic Model with Turbulent Conduction and Viscosity", 2023, The Astrophysical Journal.

- **Quantifying Temperature Probability Density Function in Turbulent Flows Through Topological Methods** (April 2024–Present)

  Applied topological methods to quantify complex interface geometry in high-resolution simulations of multi-phase astrophysical turbulence. Explained the role of turbulent mixing in setting the temperature probability density function. Results resolve the long-standing observational puzzle of a universal X-ray to Hα brightness ratio.

  Publications: **Zirui Chen** and S. Peng Oh, "From Clumps to Sheets: Geometry Controls the Temperature PDF of Multi-Phase Gas", 2026, Submitted to The Open Journal of Astrophysics.

  Zirui Chen, S. Peng Oh, et al. "Clumps in a Cocoon: Geometry and Mixing Set the Universal X-ray to Hα Surface Brightness Ratio", 2026, Submitted to Monthly Notices of the Royal Astronomical Society Letter.

- **Other Significant Publications**

  Zirui Chen, Zixuan Peng, Kate H. R. Rubin et al., "Modeling Emission-Line Surface Brightness in a Multiphase Galactic Wind: An O VI Case Study", 2026, Monthly Notices of the Royal Astronomical Society.

  Zixuan Peng, Crystal Martin, **Zirui Chen**, Drummond B. Fielding, et al., "Physical Origins of Outflowing Cold Clouds in Local Star-Forming Dwarf Galaxies", 2025, The Astrophysical Journal.

Grants & Proposals

- Hubble Space Telescope Cycle 32 Archival Research Proposal, ID 17860, **Co-PI** — October 2024

  "Interface: A New Tool for Generating Absorption and Emission Spectra from Multi-Phase Mixing Layers" . Received funding for a senior graduate student for 6 months **($50,000.00)** amid significant funding cuts at NASA.
- Non-thermal Processes at Multi-Phase Boundaries in the Interstellar and Circum-galactic Medium, ACCESS PHY240194, **Co-PI** — October 2024

  107.6k node hours and **200kGB archival storage** on the Stampede3 computing cluster at UT Austin. The estimated value of these awarded resources is **$63,800.00**.
- Turbulence and Thermally Unstable Gas, ACCESS PHY240001, **PI** — January 2024

  400k ACCESS credits on 40+ NSF supported computing clusters in the US.
- Molecular Cloud in Galactic Winds, ACCESS PHY230107, **PI** — July 2023

  400k ACCESS credits on 40+ NSF supported computing clusters in the US.

Invited Talks, Teaching, & Professional Service

- Referee for Monthly Notices of the Royal Astronomical Society.
- Invited talks at Astronomy on Tap (Astronomy Talk at a Bar), Carnegie Observatories, the Flatiron Institute (Center for Computational Astrophysics), and UCSB.
- Teaching Assistant for astronomy and experimental physics courses at UCSB (September 2022 to Present).
- Mentored undergraduate student to conduct research in computational astrophysics (January 2024 - June 2024).
- Volunteered at Columbia Astronomy Outreach Program (September 2018 - May 2022).